

Mohamad Chehade

PH.D. STUDENT · ELECTRICAL AND COMPUTER ENGINEERING

The University of Texas at Austin

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Education

The University of Texas at Austin

Austin, Texas

PH.D. IN ELECTRICAL AND COMPUTER ENGINEERING

Aug. 2023 – Present

- Advisor: Dr. Hao Zhu
- Research Interests: Large Language Model Alignment, Time Series Foundation Models, Risk-aware Transfer Learning, Learning for Optimization and Control, Neural Network Verification and Certified Robustness

The University of Texas at Austin

Austin, Texas

MASTER OF SCIENCE IN ENGINEERING

Aug. 2023 – May 2026

- Advisor: Dr. Hao Zhu
- Relevant Courses: Reinforcement Learning, Generative AI, Statistical Machine Learning, Applied Machine Learning, Probability and Stochastic Processes, Applied Stochastic Processes, Learning-based Optimal Control, Convex Optimization, Energy Optimization and Operation

American University of Beirut

Beirut, Lebanon

B.ENG. IN ELECTRICAL AND COMPUTER ENGINEERING

Aug. 2019 – Jun. 2023

- GPA: 4.0/4.0 | Minor in Mathematics
- Final Year Project: Optimal Power Flow via Machine Learning (Advisor: Dr. Rabih Jabr)
- Undergraduate Research: Microgrid Sizing using Ordinal Optimization (Advisor: Dr. Sami Karaki)

Experience

Argonne National Laboratory - Grid Optimization and Infrastructure Group

Remote

RESEARCH AIDE – MENTORS: DR. FENG QIU & DR. WEI GAO

May 2026 – Aug. 2026

- Developing decision-aware time series foundation models that align forecasting with downstream decision-making and optimization.

Los Alamos National Laboratory - T-5 Applied Mathematics and Plasma Physics Group

Remote

GRADUATE RESEARCH ASSISTANT (GRA) - MENTOR: DR. RUSSELL BENT

May 2025 - Aug. 2025

- Extended neural network verification ideas toward robustness training and certified robustness
- Developed a training algorithm that leverages verifiable input regions to improve certified robustness of neural networks

Los Alamos National Laboratory - T-5 Applied Mathematics and Plasma Physics Group

Los Alamos, NM

GRADUATE RESEARCH ASSISTANT (GRA) - MENTORS: DR. WENTING LI, DR. BRIAN BELL

Jun. 2024 - Aug. 2024

- Worked on the verification of neural networks in physical and safety-critical systems
- Developed two algorithms for determining large verifiable input regions of neural networks

University of Connecticut - Center for Clean Energy Engineering - PEARL Lab

Storrs, CT

RESEARCH INTERNSHIP - ADVISOR: DR. ALI BAZZI

Jun. 2022 - Aug. 2022

- Collaborated with Ward Leonard, a leading industrial motor manufacturer, on fault-tolerant power electronics
- Developed a fault diagnosis algorithm for power electronic inverters using combinational logic
- Optimized and constructed the inverter circuit for high-power testing using mixed-integer linear programming

Publications

PUBLISHED / ACCEPTED

Chehade, M., & Zhu, H. (2026). NEO-Grid: A Neural Approximation Framework for Optimization and Control in Distribution Grids. *Proceedings of the 59th Hawaii International Conference on System Sciences (HICSS 2026)*. arXiv:2509.21668 | Code

Chehade, M., & Zhu, H. (2026). Fine-Tuning LLMs to Generate Economical and Reliable Actions for the Power Grid. Accepted to 2026 IEEE Power & Energy Society General Meeting (PESGM 2026).

- Cho, Y.-H., **Chehade, M.**, Al-Janahi, F., Lim, S., Mohammadi, J., & Zhu, H. (2026). Carbon-aware Market Participation for Building Energy Management Systems. Accepted to 2026 *IEEE Power & Energy Society General Meeting (PESGM 2026)*.
- Chehade, M.**, Li, W., Bell, B. W., Bent, R., Kazi, S. R., & Zhu, H. (2025). LEVIS: Large Exact Verifiable Input Spaces for Neural Networks. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*. arXiv:2408.08824 | OpenReview
- Chehade, M.**, Ghosal, S. S., Chakraborty, S., Reddy, A., Manocha, D., Bedi, A. S., & Zhu, H. (2025). Bounded Rationality for LLMs: Satisficing Alignment at Inference-Time. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*. arXiv:2505.23729 | OpenReview
- Chehade, M.**, & Karaki, S. (2025). BOOST: Microgrid Sizing Using Ordinal Optimization. *2025 IEEE Texas Power and Energy Conference (TPEC)*, College Station, TX, USA, pp. 1–4. doi:10.1109/TPEC63981.2025.10907217.
- Chehade, M.**, Cho, Y.-H., Chinchali, S., & Zhu, H. (2024). Should We Use Model-Free or Model-Based Control? A Case Study of Battery Control. *2024 56th North American Power Symposium (NAPS)*, El Paso, TX, USA, pp. 1–5. doi:10.1109/NAPS61145.2024.10741791.

UNDER REVIEW

- Chehade, M.**, Bedi, A. S., Chakraborty, S., Zhang, A., & Zhu, H. (2026). Test-Time Risk Adaptation with Mixture of Agents. Under review at *NeurIPS 2026*. arXiv:2408.08812

Skills

- **Programming Languages:** Python, MATLAB, C/C++, SQL, R
- **ML / Scientific Computing:** PyTorch, NumPy, SciPy, Scikit-learn, Matplotlib, Gurobi, MATPOWER
- **Engineering & Power Systems Tools:** Simulink, SPICE, HOMER, PVSyst, LabVIEW
- **Tools & Workflow:** Git, $\text{\LaTeX}$, Linux, Overleaf, Microsoft Office Suite
- **Languages:** English (fluent), French (fluent), Arabic (native)

Reviewer

Sep. 2023 - Present	IEEE Transactions on Smart Grid
July 2024 - Present	Asilomar Conference on Signals, Systems, and Computers
Dec. 2024 - Present	Texas Power and Energy Conference (TPEC)
June 2025 - Present	Hawaii International Conference on System Sciences (HICSS)
Sep. 2025 - Present	IEEE Systems Journal
Oct. 2025 - Present	International Conference on Learning Representations (ICLR)
Feb. 2026 - Present	International Conference on Machine Learning (ICML)
May 2026 - Present	IEEE Power Engineering Letters
June 2026 - Present	Conference on Neural Information Processing Systems (NeurIPS)

Awards & Honors

2026	Third Place, ASME CIE Student Hackathon ASME IDETC/CIE 2026
2024	Best Graduate Presentation Award at NAPS 2024
2023 - Present	Cockrell School of Engineering Fellowship
2023	Mohamad Ali Safieddine Award for Academic Excellence for ranking first across the AUB Maroun Semaan Faculty of Engineering and Architecture
2023	ECE Distinguished Graduate Award for ranking first among ECE graduates
2023	Exceptional ECE Final Year Project Award Power and Energy Systems
2021	Best Pitch Award EPFL Tech4Impact Summer School
2019 - 2023	Dean's Honor List AUB Maroun Faculty of Engineering and Architecture - every given semester